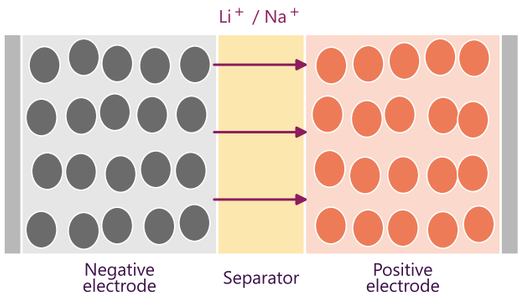
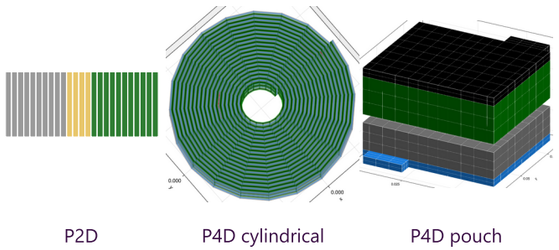
1 Physics models

- Li-ion & Na-ion batteries (DFN / P2D)
- SEI layer growth
- Thermal model & temperature dependence
- Full or reduced solid diffusion



2 Cell geometries

- P2D: fast 1D through-thickness model
- P4D: full 3D cell designs
- Cylindrical (jelly-roll) & pouch cells
- Built-in meshing via Jutul.jl



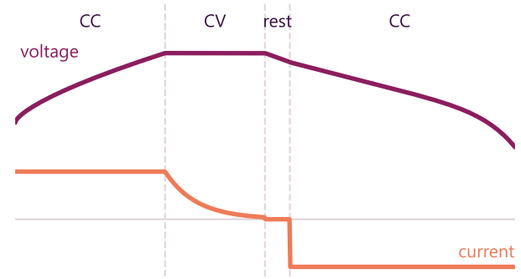
3 Inputs & parameter library

- JSON input sets, language-independent
- Default cell parameter sets & protocols
- Schema validation of all inputs
- EMMO / BattINFO ontology metadata

```
{                                          chen_2020.json
  "NegativeElectrode": {
    "ActiveMaterial": {
      "Description": "Graphite-SiOx",
      "DiffusionCoefficient": 3.3e-14,
      "ParticleRadius": 5.86e-6, ...
```

4 Cycling protocols

- CC charge / discharge
- CC-CV and multi-cycle protocols
- User-defined current functions
- Voltage limits & initial state of charge



BattMo.jl

Continuum-scale battery cell modelling in Julia

Julia API

Simulation · Calibration · solve

PyBattMo

Python wrapper

BattMoApp

Graphical web interface

Jutul.jl + Makie.jl

Solver & plotting backend

5 Fast, differentiable solver

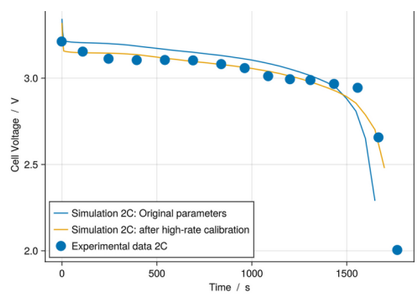
- Finite-volume discretisation on Jutul.jl
- Automatic differentiation of all equations
- Adjoint-based gradients & sensitivities
- Direct & preconditioned iterative solvers

≈ 0.5 s

for a standard P2D constant-current discharge

6 Calibration & optimisation

- Voltage calibration to experimental data
- Equilibrium (OCV) calibration
- Free / freeze parameters with bounds
- Gradient-based optimisation via adjoints



7 Cell KPIs & analysis

- Capacity, N:P ratio, mass & volume
- Mass loading & energy density
- Discharge energy, round-trip efficiency
- dQ/dV & voltage breakdown

```
quick_cell_check(cell)
```
design KPIs before simulating

```
get_output_metrics(output)
```
performance KPIs after simulating

```
compute_dqdv(q, V)
```
differential capacity analysis

8 Visualisation

- One-line result dashboards
- Time series & spatial contour plots
- Interactive 3D plots of P4D results
- Ontology-labelled outputs & units

